Advancing Image Inpainting: Techniques for Seamless Visual Restoration

Hassan Naeem
Department of Artificial Intelligence
National University of Computing and Emerging Sciences
Islamabad, Pakistan
i210284@nu.edu.pk

Jannat Butt
Department of Artificial Intelligence
National University of Computing and Emerging Sciences
Islamabad, Pakistan
i210259@nu.edu.pk

Manahil Sarwar
Department of Artificial Intelligence
National University of Computing and Emerging Sciences
Islamabad, Pakistan
i210293@nu.edu.pk

***Abstract*—Image Inpainting is referred to filling the parts of image that are not present or in other words restoration of an image. This paper proposes advanced techiniques for image inpainting using different types of CNN and GANS. The work is done on CIFAR-10 dataset, by first removing part of the images and then using those to train the models for image inpainting. The work proposed makes great progress in the field of image inpainting.**

Keywords: Image Inpainting, Image Restoration, Image Manipulation, Deep Learning

I. Introduction

Generative AI has gained significant popularity in the recent years due its great capabilities. There is a lot of work done on image generation that has wowed everyone. From text generation to image to now work being done on video generation on prompt, generative AI has brought a revolution in the field of AI and has worked great in improving quality of life. Needless to say it still currently is under rapid progression and keeps improving day by day. This paper talks proposes work done on one of the many issues generative AI addresses and that is Image inpainting. Image inpainting refers to the restoration of images by filling out missing parts of an image or generally improving the quality of an original image. It has been under work for quite some time and only recently there are many great ways developed that one can use to perform image restoration. The work done on image inpainting involves purposefully destroying images to then perform restoration on them. 10 type of different models were trained out of which 5 worked exceptionally while the other performed fairly poor and failed to converge. The 10 models that were trained contain some written from scratch while some are pretrained and finetuned on our specific dataset to perform image restoration. The primary contributions of this work are summarized as follows:

- Utilizing GANS, CNNs, Encoders and U-NET and their variations a few models are trained and tested to perform seamless and high fidel image restoration.
- CIFAR-10 Dataset is used for training and evaluation of all the models used.
- Image reconstructed are consistent and clear.
- Different experiments and methods are used for evaluation of all the models resulting in a thorough examination of the working, shortcomings and pluspoints of each model. This paper is contains information on previous work done on Image inpainting, information on dataset used for training and evaluation, Methodology used and experimental setups and results.

II. Related Work

Summary related to a few of research papers proposing work on image inpainting is as follows:

A. Image Inpainting for Corrupted Images Using the Semi-Super Resolution GAN

Citation: Tayefeh, M. M., Tayefeh, M. M., & Ghahramanie, S. A. G. (2024). Image inpainting for corrupted images using the semi-super resolution GAN. Sharif University of Technology.

Research Question: The paper addresses restoring corrupted images using deep learning to recover missing pixels.

Methods: Semi-Super Resolution GAN (SSRGAN) is used for inpainting by predicting missing pixels.

Datasets: OxfordIIITPet, Caltech101, Flower102.

Accuracy: Normalized Mean Squared Error (NMSE).

Contribution: SSRGAN improves image restoration by scaling inpainting for diverse levels of corruption.

B. An Image Inpainting Method Based on Generative Adversarial Networks Inversion and Autoencoder

Citation: Wang, Y., Song, B., & Zhang, Z. (2024). An image inpainting method based on generative adversarial networks inversion and autoencoder. *IET Image Processing, 18*(4), 1042–1052.

Research Question: Inpainting large-area, high-resolution damaged images.

Methods: GAN-based autoencoder for feature extraction and GAN inversion.

Datasets: CelebAMask-HQ, Flickr-Faces-HQ, ImageNet.

Accuracy: Structural and texture accuracy.

Contribution: Enhanced image inpainting for large-area damage with superior realism.

C. Locate, Assign, Refine: Customized Image Inpainting with Text-Subject Guidance

Citation: Zhang et al. (2024). *ArXiv*.

Research Question: Precise reconstruction of subject-specific regions in complex scenes.

Methods: Three-step approach: Locate, Assign, and Refine, with decoupled cross-attention.

Dataset: Custom inpainting datasets.

Evaluation: PSNR, SSIM.

Contribution: Improved control over subject identity in inpainted regions while maintaining image fidelity.

D. Diffusion-Based Image Inpainting with Internal Learning

Citation: Portenier, T., Gleyzes, J.-B., & Tschannen, M. (2023). Diffusion-based image inpainting with internal learning (*arXiv:2406.04206v1*).

Research Question: Inpainting via internal learning without external datasets.

Methods: UNet-based diffusion model with multi-scale adaptive receptive field.

Datasets: Texture dataset, line drawing dataset, SVBRDF material dataset.

Accuracy: PSNR, SSIM, LPIPS, MSE.

Contribution: Efficient inpainting with reduced dependence on external data.

E. Image Inpainting for Irregular Holes Based on Generative Adversarial Networks

Citation: Liu, T., Gao, Z., & Li, W. (2024). Image inpainting for irregular holes based on generative adversarial networks. *SPIE 13107, Fourth International Conference on Sensors and Information Technology*.

Research Question: Improving image inpainting for irregular masks.

Methods: Three-stage restoration network with multi-scale discriminators.

Datasets: CelebA-HQ, Places2, Paris StreetView, QD-IMD.

Accuracy: PSNR, SSIM, L1 loss, FID.

Contribution: Reducing structural deviations and boundary artifacts in inpainting.

F. Image Inpainting Using Attention Mechanisms and Convolutional Neural Networks

Citation: Guo, Y., Yu, S., Zhang, Z., Wang, J. (2023). Image Inpainting Using Attention Mechanisms and Convolutional Neural Networks. Applied Sciences, 13(20), 11189. https://doi.org/10.3390/app132011189

Research Question: How to enhance image inpainting accuracy and efficiency using attention mechanisms combined with convolutional neural networks.

Methods: Combines attention-based features with convolutional neural networks to focus on important regions for inpainting, ensuring detailed restoration.

Contribution: Developed a framework that improves image completion results by accurately reconstructing complex textures and structures.

G. Restore from Restored: Single-image Inpainting

Citation: Lee, E., Kim, J., Kim, J., Kim, T. H. (2021). Restore from restored: Single-image inpainting. arXiv preprint arXiv:2110.12822.

Research Question: Addressing the limitations of existing single-image inpainting methods for more effective image restoration.

Methods: The approach improves the image step by step, fixing it multiple times to make the result look more natural and clear.

Contribution: This method creates better results with fewer mistakes making the fixed images look much closer to real ones.

H. Image Inpainting and Restoration Based on Variational and Convolution Techniques

Citation: Dinu, V., Ionescu, F., Popa, A. M. (2023). Image inpainting and restoration. ResearchGate Publication.

Research Question: Enhancing the quality of image inpainting by leveraging variationall techniques and convolutional methodologies.

Methods: Utilizes aa combination of Bertaalmio's algorithm with anisotropicc diffusion techniques for smooth propagation.

Contribution: Introduced a nice framework interleeaving inpainting and diffusion processes.

I. Image inpainting: A review

Citation: Elharrouss, O., Almaadeed, N., Al-Maadeed, S., Akbari, Y. (2019). Image inpainting: A review. arXiv preprint arXiv:1909.06399.

Contribution: This is a contribution for digital image inpainting researchers trying to look for the available datasets because there is a lack of datasets available for image inpainting.

III. Data Set

This paper utilizes a well known dataset named as CIFAR-10 dataset which contains 60,000 categorized images into 10 classes. Each category has 6000 images. The categories are airplane,automobile,bird,cat,deer, dog,frog,horse,ship,truck. A sample of data is :

Fig. 1. Dataset

A. Data Preprocessing

To make the data ready for image inpainting we a masking function which adds black squares to random areas of the image. This makes missing parts of an image, helping the model learn how to fill in the gaps. Along with this a series of image transformations and augmentations were applied to the data. Data after inpainting:

Fig. 2. Masked Images.

B. Data Distribution

The Dataset was split into training, testing and validatiom data. The dataset distributions were as follows:

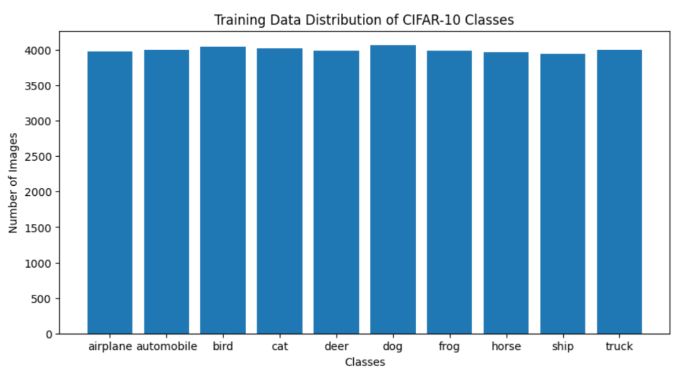

Fig. 3. Training Distribution.

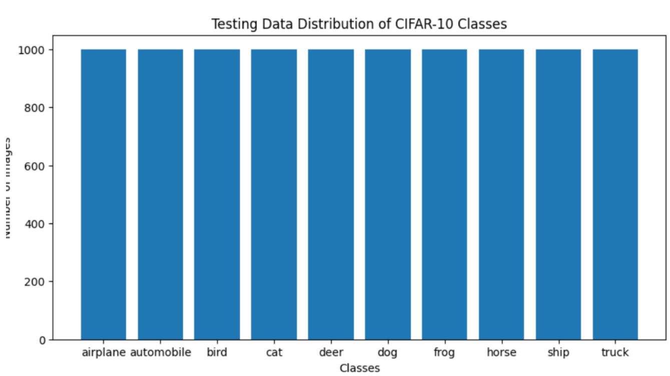

Fig. 4. Testing Distribution.

IV. Methodology

The methodology used in this paper consisted of using 5 models to train on our CIFAR-10 masked images to use for image inpainting. The 5 models were :

A. Pixel CNN

We implemented a Pixel-CNN based model for image inpainiting. It contains convolutional base with residual layers which predicts RGB logits for pixel values in masked regions. This model was trained using Cross-entropy loss along with early stopping and model checkpointing. The model was evaluated using loss over epochs and PSNR values. The training and validation loss of the model was plotted as follows:

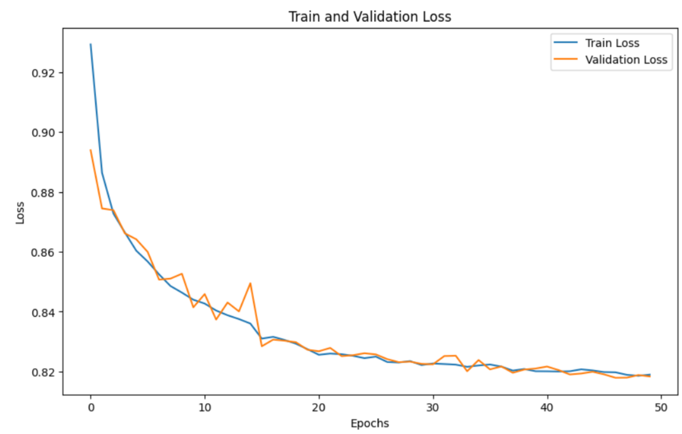

Fig. 5. training and validation loss

1) Experimental Results: The PSNR and average test loss of the model was : Some visualized results are as follows:

```
Final Test Loss: 0.8210
Average PSNR: 29.86 dB
```

Fig. 6. The PSNR and average test loss

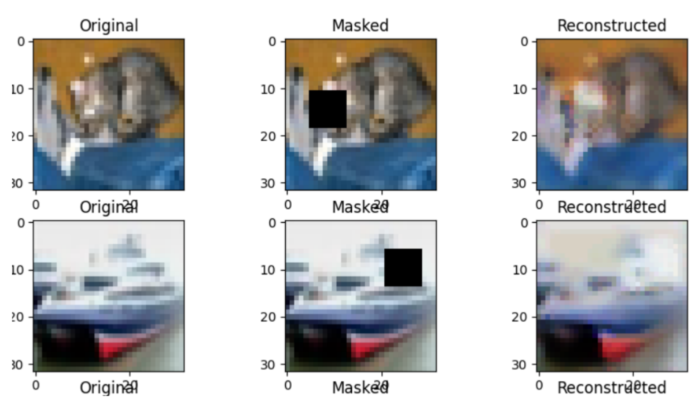

Fig. 7. visualized results

2) Limitations: The PixelCNN-based model for image inpainnting is slow for high-resolution images because it processes one pixel. It also has difficulty filling in large or complex missing areas consistently.

B. Autoencoder for Image Inpainting

We implemented a Autoencoder for Image Inpainting. The autoencoder consists of two main components:
1) Encoder: Reduces the spatial dimensions of the image while extracting feature representations. It uses convolutional layers, ReLU activations, and max-pooling to downsample the input image into featre-rcih smaller representations.
2) Decoder: Reconstructs the original image from the encoded features. Uses convolutional layers, ReLU activations, and upsaampling layers to expanad the spatial dimensiions back to the original image size.
The final layer applies the Tanh activation to ensure the output values are normalized between -1 and 1.
This model was trained using Mean Squared Error (MSE) loss along with A learning rate scheduler (ReduceLROnPlateau) and Early Stopping. The model was evaluated using reconstruction loss over epochs and PSNR values. The training and validation loss of the model was plotted as follows:

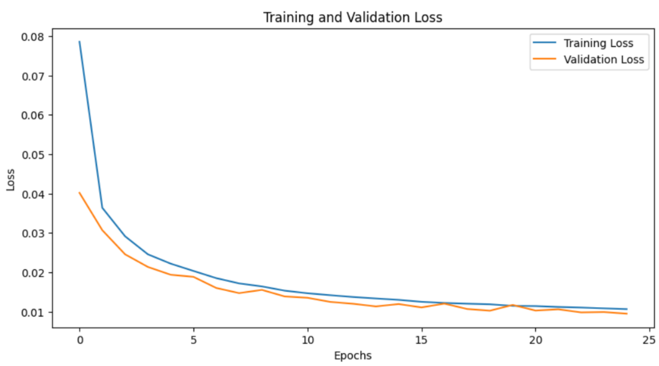

Fig. 8. training and validation loss

1) Experimental Results: The PSNR and average test loss of the model was : Some visualized results are as follows:

```
Average Reconstruction Loss: 0.0095
Average PSNR: 20.21 dB
```

Fig. 9. The PSNR and average test loss

Fig. 10. visualized results

2) Limitations: The inpainting autoencoder is limited by its inability to capture complex globbal structures and fine details in big misssing areas as it focuses on local features more than global features. It also causes image downsampling that results in not a very clear picture.

C. Partial Convolution-based GAN

Partial Convolution-based GAN implemetaed contain:
Generator:

The generator uses partial convolutions to handles irregular masks, ensuring the masked region is updated while keeping unmasked regions intact. Discriminator:
It is a PatchGAN-style discriminator that determine whether the image patches are real or generated (fake). It processes image through a series of convolutional layer with LeakyReLU activation and Batch Normalization, culminating in a Sigmoid output for binary classification.
This model was trained using Reconstruction Loss and Adversarial Loss along with A learning rate scheduler (ReduceLROnPlateau) and Early Stopping. The model was evaluated using reconstruction loss over epochs and PSNR values.

1) Experimental Results: The PSNR and average test loss of the model was : Some visualized results are as follows:

```
Average Reconstruction Loss: 0.0633
Average PSNR: 20.7528
```

Fig. 11. The PSNR and average test loss

Fig. 12. visualized results

2) Limitations: The generator may struggles with complex or large masked regions. While partial convolution improve handling irregular masks, their effectiveness diminishes when most of the image is occluded. Despite partial convolutions, artifacts (e.g., blurry or incomplete reconstruction) may appears in the generated outputs, particularly at the boundaries of masked regions. The current architecture does not explicitly incorporates multi-scale context, which is critical for reconstructing large missing area or fine details.

D. Vanilla GAN with Encoder-Decoder Generator

The Vanilla GAN with Encoder-Decoder Generator implemetaed contain:
Generator:
The generator uses an encoder-decoder architecture. It learns to inpaint or reconstructing masked regions in images.
The encoder compress the input to a latent representation, and the decoder reconstruct the image.
The generator's output is a new image with reconstructed regions, where unmasked parts are ideally unchanged.
Discriminator:
The discriminator is a classifier that decides whether an image is real (from the dataset) or fake (created by the generator).
It uses a series of convolutional layers with LeakyReLU activations for extracting features and ultimately outputs a probability value.
This model was trained using Reconstruction Loss and Adversarial Loss along with A learning rate scheduler (ReduceLROnPlateau) and Early Stopping. The model was evaluated using reconstruction loss over epochs and PSNR values. The training and validation loss of the model was plotted as follows:

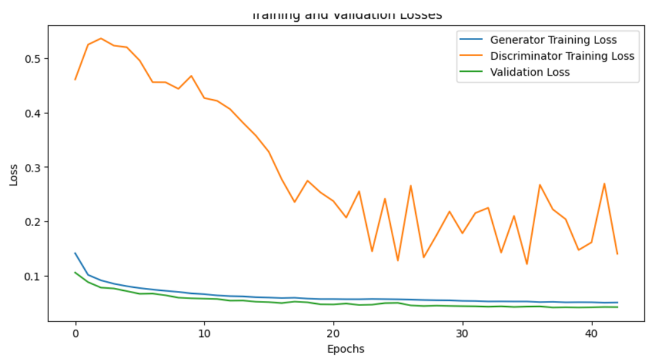

Fig. 13. training and validation loss

1) Experimental Results: The PSNR and average test loss of the model was : Some visualized results are as follows:

```
Average Reconstruction Loss (L1): 0.0512
PSNR Score: 18.87 dB
```

Fig. 14. The PSNR and average test loss

Fig. 15. visualized results

2) Limitations: GAN training is known to be unstable due to the adversarial setup, where the generator and discriminator trying to outperform each other. GANs are also computationally expensive to train becuase of the iterative nature of updates and the extra adversarial loss component. The generator might perform well on training data but fails to generalize to unseen masked images.

V. Conclusion

In conclusion, we have explored information on previous work done on Image inpainting, information on dataset used for training and evaluation, Methodology used and experimental setups and results along with different models used in the process and limitations of each of them.

References

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